

Introduction to Kali Linux

Basic Commands

I executed several basic Linux commands in Kali Linux:

whoami

hostnamectl

uname -a

df -h

ip a

The commands displayed information about:

- logged in user
- operating system
- kernel version
- disk usage
- network configuration

```

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$ hostnamectl
Static hostname: kali
Icon name: computer-vm
Chassis: vm
Machine ID: 686fa41363f049e9a84be345a7078e04
Boot ID: c9b929521f4b450e933b54369a7ffc3c
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 6.12.38+kali-amd64
Architecture: x86-64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 5month 3w 2d

(kali@kali)-[~]
$ uname -a
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64 GNU/Linux

(kali@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            921M   0    921M   0% /dev
tmpfs           198M  976K  197M   1% /run
/dev/sda1       79G   16G   59G   21% /
tmpfs           987M   4.0K  987M   1% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           987M   8.0K  987M   1% /tmp
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  120K  198M   1% /run/user/1000

(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1f:b7:23 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86238sec preferred_lft 86238sec
    inet6 fd17:625c:f037:2:1870:9c19:d68c:274f/64 scope global dynamic noprefixroute
        valid_lft 86241sec preferred_lft 14241sec

```

HTOP

I installed and tested the htop package.

```
sudo apt install htop -y
```

htop

HTOP displays running processes and system resource usage in real time.

```

0 [|||||] 0.00 Tasks: 87, 248 thr, 77 kthr; 1 running
1 [|||||] 6.7% Load average: 0.32 0.46 0.74
Mem[|||||] 69M/2.0G Uptime: 00:05:37
Swap[|||||] 0K/95M

Main [|||||]
PID USER PRI NI VIRT RES PRIV VS CPU/MEM TIME+ Command
783 root 20 0 431M 139M 5528 5 3.4 0.6 0:07:24 /usr/lib/xorg/xorg -o -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -metswitch
22 root 20 0 441M 135M 5404 5 2.4 0.5 0:04:34 /usr/lib/xorg/xorg -o -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -metswitch
1835 kali 20 0 642M 690M 15732 5 2.1 1.4 0:01:04 /usr/bin/gdmflexiblerenderd -o -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -metswitch
3511 kali 20 0 965M 546M 1920 1.4 0.3 0:00:27 http
1850 kali 20 0 965M 120M 42676 0 0 0.6 0:01:36 fdm
1113 kali 20 0 965M 262M 7184 5 0 0.7 0:00:67 /usr/lib/x86_64-linux-gnu/xfce4/pa/xfce4-panel -o /usr/lib/x86_64-linux-gnu/xfce4/pa/plugins/libgdmemo.so.15 231659838 gdmemo Gnome Monitor Show output of a command
1837 kali 20 0 642M 690M 1 0 0 0.7 0:00:65 /usr/bin/gdmflexiblerenderd
1883 kali 20 0 642M 690M 1 0 0 0.7 0:00:34 /usr/bin/gdmflexiblerenderd
1 root 20 0 2355M 1408M 3520 5 0 0.7 0:00:96 /sbin/init splash
343 root 20 0 518M 167M 1280 5 0 0.8 0:00:17 /usr/lib/systemd/systemd-journald
394 root 20 0 3651M 18092 2816 5 0 0.5 0:00:12 /usr/lib/systemd/systemd-udev
514 root 20 0 538M 542M 4880 5 0 0.3 0:00:17 /usr/sbin/avahi-daemon --foreground --verbose=1
516 root 20 0 301M 122M 768 5 0 0.4 0:00:07 /usr/libexec/accounts-daemon
517 messages 20 0 1026M 723M 3072 0 0 0.4 0:00:42 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
518 polkitd 20 0 373M 18272 2836 5 0 0.5 0:00:16 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
519 root 20 0 1902M 9432 1280 5 0 0.5 0:00:16 /usr/lib/systemd/systemd-logind
557 root 20 0 688M 268M 128 5 0 0.1 0:00:00 /usr/sbin/cron -f
574 root 20 0 301M 722M 0 0 0 0.5 0:00:00 /usr/libexec/accounts-daemon
575 root 20 0 301M 722M 0 0 0 0.4 0:00:00 /usr/libexec/accounts-daemon
580 polkitd 20 0 373M 18272 0 0 0 0.5 0:00:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
597 polkitd 20 0 373M 18272 0 0 0 0.5 0:00:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
609 root 20 0 320M 1932M 2944 5 0 1.0 0:00:20 /usr/sbin/NetworkManager --no-daemon
610 polkitd 20 0 373M 18272 0 0 0 0.5 0:00:04 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
611 root 20 0 301M 722M 0 0 0 0.5 0:00:01 /usr/libexec/accounts-daemon
621 root 20 0 309M 1536 1920 5 0 0.6 0:00:13 /usr/sbin/ModemManager
628 root 20 0 204M 337M 264 5 0 0.2 0:00:00 /usr/sbin/WiXService
630 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
631 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
632 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
633 root 20 0 204M 337M 0 0 0 0.2 0:00:07 /usr/sbin/WiXService
634 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
635 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
636 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
637 root 20 0 204M 337M 0 0 0 0.2 0:00:00 /usr/sbin/WiXService
638 root 20 0 309M 1536 0 0 0 0.6 0:00:00 /usr/sbin/ModemManager
640 root 20 0 309M 1536 0 0 0 0.6 0:00:00 /usr/sbin/ModemManager
642 root 20 0 309M 1536 0 0 0 0.6 0:00:00 /usr/sbin/ModemManager
643 root 20 0 320M 1932M 0 0 0 1.0 0:00:00 /usr/sbin/NetworkManager --no-daemon
644 root 20 0 320M 1932M 0 0 0 1.0 0:00:00 /usr/sbin/NetworkManager --no-daemon
645 root 20 0 320M 1932M 0 0 0 1.0 0:00:00 /usr/sbin/NetworkManager --no-daemon
688 root 20 0 372M 730M 768 5 0 0.4 0:00:01 /usr/sbin/lightdm
694 root 20 0 372M 730M 0 0 0 0.4 0:00:00 /usr/sbin/lightdm
695 root 20 0 372M 730M 0 0 0 0.4 0:00:00 /usr/sbin/lightdm
697 root 20 0 372M 730M 0 0 0 0.4 0:00:00 /usr/sbin/lightdm

```

Traceroute

I tested network diagnostics using traceroute.

tracert google.com

The command showed the route packets take through the network.

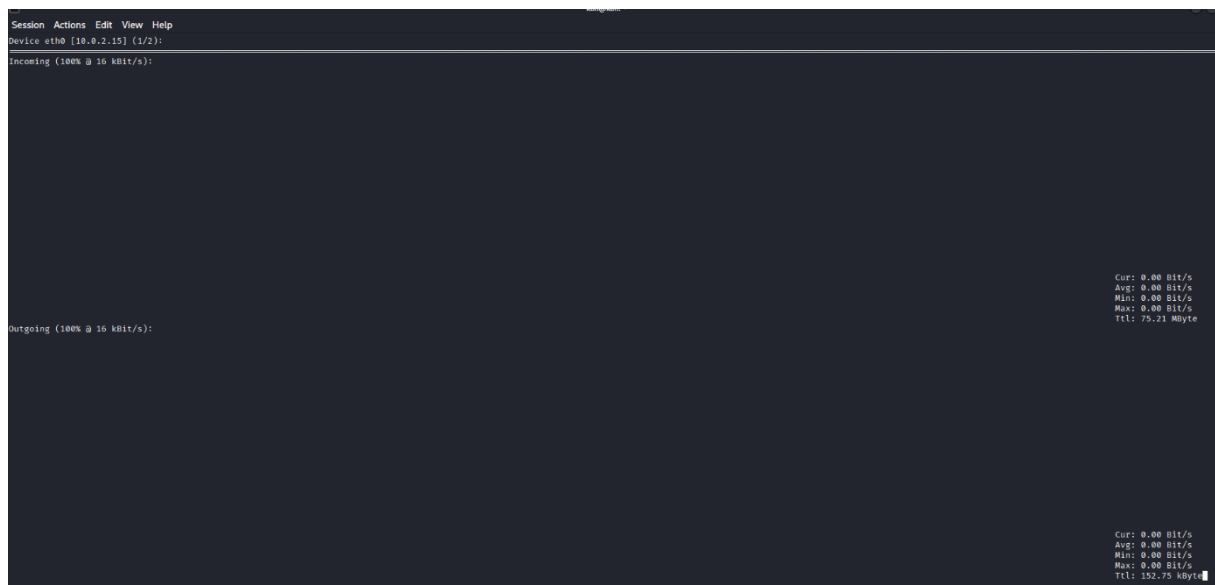
```
(kali㉿kali)-[~]
└─$ traceroute google.com
traceroute to google.com (172.217.168.142), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  5.917 ms  5.287 ms  4.801 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Nload

I tested the nload tool for monitoring network traffic.

nload

The tool graphically displayed incoming and outgoing traffic.



Searchsploit

I used Searchsploit to search for WordPress FTP vulnerabilities.

searchsploit wordpress ftp

The tool displayed several known vulnerabilities and exploits.

kali@kali:~\$ searchsploit wordpress ftp	
Exploit Title	Path
WordPress Plugin MimiFTP 1.0.5 - Arbitrary File Download (1)	php/webapps/36774.txt
WordPress Plugin MimiFTP 1.0.5 - Arbitrary File Download (2)	php/webapps/36881.txt
WordPress Plugin MimiFTP 1.0.5 - Cross-Site Request Forgery / Arbitrary File Creation / Remote Code Execution	php/webapps/36763.txt
WordPress Plugin MimiFTP 1.0.5 - Cross-Site Request Forgery / Arbitrary File Deletion	php/webapps/36761.txt
WordPress Plugin MimiFTP 1.0.5 - Multiple Cross-Site Request Forgery / Cross-Site Scripting Vulnerabilities	php/webapps/36762.txt

DNSenum

I tested DNS enumeration on the google.com domain.

dnsenum google.com

The tool displayed DNS records, mail servers, and nameservers.

```
dnsenum VERSION:1.3.1

— google.com —

Host's addresses:
-----
google.com.                234      IN      A       172.217.168.142

Name Servers:
-----
ns3.google.com.            85992    IN      A       216.239.36.10
ns2.google.com.            15121    IN      A       216.239.34.10
ns1.google.com.            15121    IN      A       216.239.32.10
ns4.google.com.            85992    IN      A       216.239.38.10

Mail (MX) Servers:
-----
smtp.google.com.           300      IN      A       64.233.184.26
smtp.google.com.           300      IN      A       66.102.1.27
smtp.google.com.           300      IN      A       66.102.1.26
smtp.google.com.           300      IN      A       142.250.110.26
smtp.google.com.           300      IN      A       64.233.184.27

Trying Zone Transfers and getting Bind Versions:
-----

Trying Zone Transfer for google.com on ns4.google.com ...
AXFR record query failed: REFUSED

Trying Zone Transfer for google.com on ns3.google.com ...
AXFR record query failed: REFUSED

Trying Zone Transfer for google.com on ns1.google.com ...
AXFR record query failed: REFUSED

Trying Zone Transfer for google.com on ns2.google.com ...
AXFR record query failed: REFUSED

Brute forcing with /usr/share/dnsenum/dns.txt:
-----
```

Name-That-Hash

I used the Name-That-Hash tool to identify a hash value.

```
nth -t ef487f75307f96954d3bb132e5f4b035
```

The tool identified the hash as most likely MD5/NTLM.

```
kali@kali:~$ nth -t ef487f75307f96954d3bb132e5f4b035

Name-That-Hash
https://twitter.com/hee_sec_son
https://github.com/hee_sec_son/Name-That-Hash

ef487f75307f96954d3bb132e5f4b035

Most Likely
MD5, HC: 0 JTR: raw-md5 Summary: Used for Linux Shadow files.
MD4, HC: 900 JTR: raw-md4
NTLM, HC: 1000 JTR: nt Summary: Often used in Windows Active Directory.
Domain Cached Credentials, HC: 1100 JTR: mscach

Least Likely
Domain Cached Credentials 2, HC: 2100 JTR: mscach2 Double MD5, HC: 2600 Tiger-128, Skein-256(128), Skein-512(128), Lotus Notes/Domino 5, HC: 8600 JTR: lotus5 md5(md5(md5($pass))), HC: 3500 Summary: Hashcat mode is only supported in
hashcat-legacy. md5(uppercase(md5($pass))), HC: 4300 md5(sha1($pass)), HC: 4400 md5(utf16($pass)), JTR: dynamic_29 md4(utf16($pass)), JTR: dynamic_33 md5(md4($pass)), JTR: dynamic_34 Haval-128, JTR: haval-128-4 RIPEMD-128, JTR:
ripemd-128 MD2, JTR: md2 Snefru-128, JTR: snefru-128 DNSSEC(KSEC3), HC: 8300 RAdmin v2.x, HC: 9900 JTR: radmin Cisco Type 7, BigCrypt, JTR: bigcrypt

kali@kali:~$
```

Reflection

Kali Linux is useful for cybersecurity testing because it contains many pre-installed security tools. The most interesting tools for me were HTOP, Searchsploit, and DNSenum.